

Astronomer Enterprise Airflow – 12 Days (4 Hours Per Day)

Day 1 – Airflow Foundations & Core Architecture Refresher

Module 1 – Airflow Architecture Overview

Topics

- Evolution of workflow orchestration
- Airflow ecosystem overview
- Core Airflow components
- Scheduler
- Webserver
- Workers
- Metadata database
- Message queues
- DAG lifecycle

Hands-on

- Install Airflow locally
- Run example DAG
- Explore Airflow UI
- Inspect DAG execution flow

Module 2 – Airflow Core Internals (Deep Dive)

Topics

- Scheduler architecture
- DAG parsing mechanism
- Task lifecycle
- Task states
- Executor models
- Metadata database schema
- Airflow configuration structure

Hands-on

- Explore metadata database tables
- Inspect scheduler logs
- Trace task lifecycle from creation to completion

Day 2 – Airflow Executors & Execution Architecture

Module 1 – Executor Models

Topics

- Sequential Executor
- Local Executor
- Celery Executor
- Kubernetes Executor
- Hybrid execution models
- Worker scaling patterns

Hands-on

- Configure Local Executor
- Execute parallel tasks
- Analyze resource usage

Module 2 – Performance and Scheduling Mechanics

Topics

- DAG scheduling algorithms
- Parallelism configuration
- DAG concurrency settings
- Queue management
- Worker autoscaling

Hands-on

- Modify Airflow configuration
- Test scheduler throughput
- Simulate high-volume scheduling

Day 3 – Astronomer Platform Architecture

Module 1 – Astronomer Managed Airflow

Topics

- Why Astronomer for enterprise
- Managed Airflow architecture
- Control plane vs data plane
- Deployment architecture
- Worker scaling model

- Enterprise orchestration use cases

Module 2 – Astro CLI Fundamentals

Topics

- Astro CLI overview
- Project structure
- Local Airflow environment
- Image build lifecycle

Hands-on

- Install Astro CLI
- Initialize Astro project
- Run Airflow locally
- Deploy first DAG

Day 4 – Astronomer Deployment Lifecycle

Module 1 – Deployment Environments

Topics

- Dev vs Stage vs Production environments
- Deployment workflow
- Docker image registry lifecycle
- Environment variables
- Connection configuration

Hands-on

- Deploy DAG to Astro
- Trigger DAG runs
- Inspect logs
- Debug deployment errors

Module 2 – Secrets and Configuration Management

Topics

- Secrets management models
- Connections
- Variables
- Secure credential storage
- Governance principles

Hands-on

- Configure secrets
- Setup secure connections
- Implement RBAC roles

Day 5 – Airflow Development Best Practices

Module 1 – DAG Design Best Practices

Topics

- Modular DAG design
- DAG maintainability
- Dependency design
- Task grouping
- Code structure standards

Hands-on

- Refactor poorly designed DAG
- Implement modular DAG

Module 2 – Task Reliability and Failure Handling

Topics

- Task idempotency
- Retry strategies
- Failure recovery patterns
- Circuit breaker patterns
- Timeout handling

Hands-on

- Implement retry patterns
- Simulate task failures
- Build resilient DAG

Day 6 – Backfill Strategies & DAG Lifecycle Management

Module 1 – Backfill Strategies

Topics

- Historical data processing
- Backfill vs catchup
- Data consistency considerations
- Operational impact

Hands-on

- Execute backfill job
- Compare catchup behaviors

Module 2 – DAG Versioning and Governance

Topics

- DAG lifecycle management
- Version control strategies
- Code promotion across environments
- Governance practices

Hands-on

- Implement DAG versioning
- Promote DAG across environments

Day 7 – DAG Factory & Config Driven DAGs

Module 1 – Introduction to DAG Factory

Topics

- Configuration driven DAGs
- YAML based DAG definitions
- Enterprise DAG templating
- Reusability patterns

Module 2 – Building DAGs Using DAG Factory

Hands-on

- Convert Python DAG to YAML DAG

- Generate multiple DAGs
- Deploy DAG Factory pipelines

Day 8 – Remote Execution Agents & Hybrid Execution

Module 1 – Remote Execution Architecture

Topics

- Hybrid orchestration
- Remote execution agents
- On-prem + cloud orchestration
- Security model

Module 2 – Implementing Remote Execution Agents

Hands-on

- Setup Kubernetes cluster
- Install execution agents
- Connect agent to Astro
- Validate task execution

Day 9 – CI/CD & DevOps Integration

Module 1 – Airflow CI/CD Architecture

Topics

- Git based DAG deployment
- CI/CD pipelines for Airflow
- Image build pipelines
- Automated testing

Hands-on

- Implement GitHub Actions pipeline
- Automate DAG deployment

Module 2 – Infrastructure Automation

Topics

- Terraform for Astronomer
- Environment provisioning
- Governance model

Hands-on

- Deploy Airflow environment using Terraform

Day 10 – Observability & Production Monitoring

Module 1 – Airflow Monitoring

Topics

- Metrics collection
- SLA monitoring
- Alerting mechanisms
- Logging architecture

Hands-on

- Configure SLA monitoring
- Simulate alert conditions

Module 2 – Astro Observe

Topics

- Orchestration-native observability
- DAG performance analysis
- Lineage visibility
- Data-aware monitoring

Hands-on

- Enable Astro Observe
- Analyze historical DAG performance

Day 11 – Disaster Recovery & Platform Upgrades

Module 1 – Disaster Recovery Strategy

Topics

- Airflow backup strategy
- Metadata database backup
- DAG repository backup
- Multi-region deployment patterns

Hands-on

- Backup metadata database
- Restore Airflow environment
- Test DR workflow

Module 2 – Rollback and Recovery

Topics

- DAG rollback strategies
- Deployment rollback
- Failure recovery workflows
- Incident response playbooks

Hands-on

- Rollback failed deployment
- Recover DAG execution

Day 12 – Airflow Upgrade Strategy & Capstone

Module 1 – Airflow Upgrade Planning

Topics

- Airflow version upgrade strategy
- Dependency compatibility management
- Plugin compatibility
- Migration testing procedures

Hands-on

- Upgrade Airflow version in test environment
- Validate DAG compatibility

Module 2 – Capstone Project

Participants will implement a **complete enterprise orchestration workflow**

Activities

1. Develop enterprise DAG using DAG Factory
2. Deploy via CI/CD pipeline
3. Configure remote execution agents
4. Implement monitoring with Astro Observe
5. Simulate failure scenario
6. Execute rollback and recovery
7. Validate DR procedures